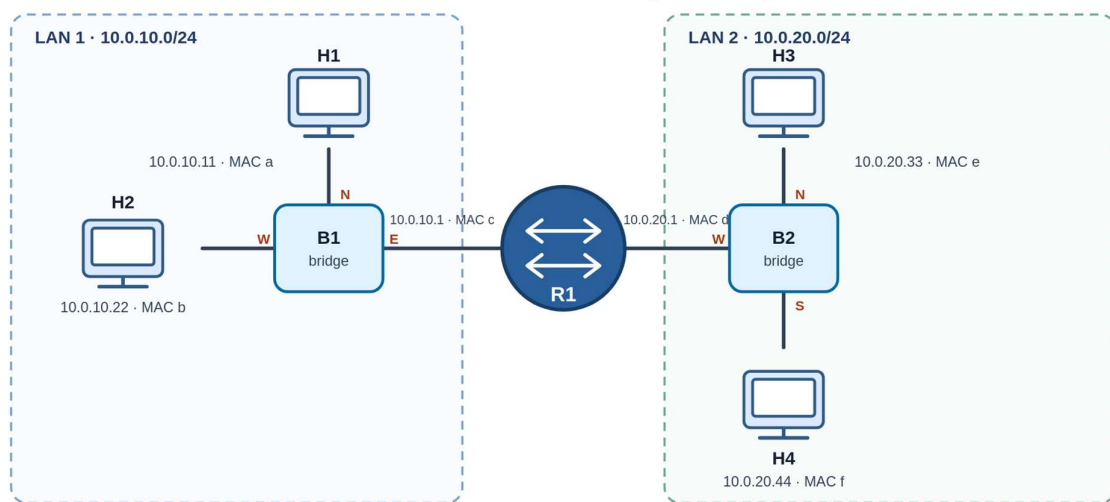


EP2120 Internetworking IK2218 Protocols and Principles of the Internet

Homework Assignment 2 (Solutions due 19:00, Mon., 14 Sep 2026) (Review due 19:00, Wed., 16 Sep 2026)

1. ARP (20/100)



The figure above shows four hosts, H1, H2, H3, and H4, connected through two learning bridges and one router.

Bridge B1 connects H1, H2, and the left interface of router R1. Bridge B2 connects H3, H4, and the right interface of R1. The IP and MAC addresses are shown in the figure. All subnets use a /24 subnet mask. The bridge ports are named N, E, W, and S.

Assume that the ARP caches of the routers and of the hosts, and the MAC address tables of the bridges are initially empty and that no packets have been sent yet. The forwarding tables of all hosts and routers are correctly configured. All hosts know the IP addresses of each other. ARP snooping (also called passive ARP learning) is enabled.

Consider that host H₁ sends an IPv4 unicast datagram to host H₄.

- Provide the state of the ARP caches of the hosts and routers as they will appear after the IPv4 unicast datagram has been delivered to host H₄, that is, after dynamic ARP resolution has been made (12p).
- Provide the state of the MAC tables of bridges B₁ and B₂ as they will appear after the IPv4 unicast datagram has been delivered to host H₄, that is, after dynamic ARP resolution has been made. (4p)

- c) Assume now that ARP snooping was disabled from the beginning on all the hosts and routers. Provide the state of the ARP caches of the hosts and routers as they would appear after the IPv4 unicast datagram has been delivered to host H₄, that is, after dynamic ARP resolution has been made (4p).

2. UDP and fragmentation (15/100)

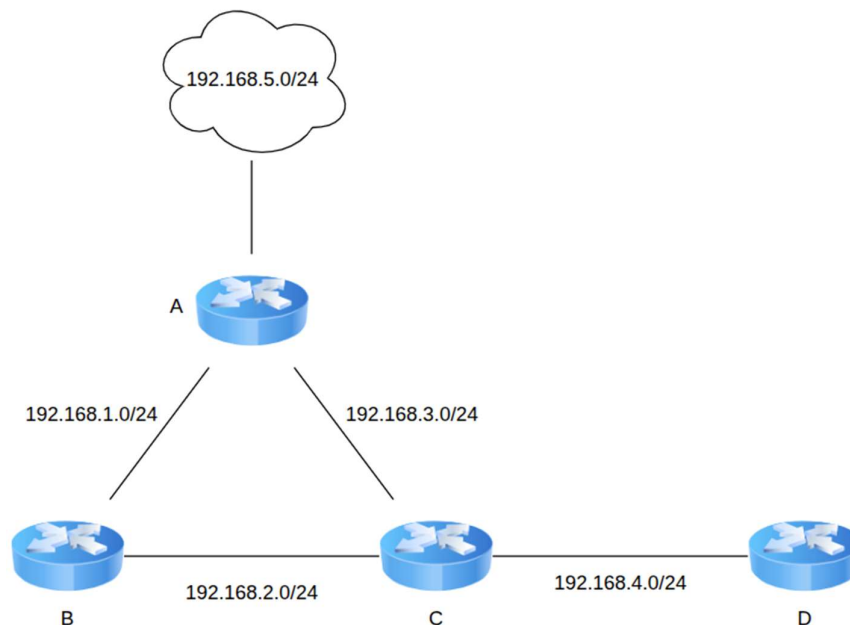
Assume that an Ethernet network with an MTU of 1280 bytes connects hosts A and B. An application process on Host A sends 5400 bytes of application data via UDP to a process on Host B. IPv4 is used at the network layer. IP options are not used.

- a) How many fragments are transmitted? (5p)
- b) Give the values of the MF bit, the offset and the total length field of the IP header of each fragment. (10p)

3. Routing (15/100)

Routers A, B, C, and D run RIPv2. Every link has metric 1 and every network has a prefix length of 24. The networks are as follows:

Network	Address Block	Attached routers	Interface Addresses
net1	192.168.1.0/24	A, B	A=.1, B=.2
net2	192.168.2.0/24	B, C	B=.1, C=.2
net3	192.168.3.0/24	A, C	A=.1, C=.2
net4	192.168.4.0/24	C, D	C=.1, D=.2
net5	192.168.5.0/24	A	A=.1



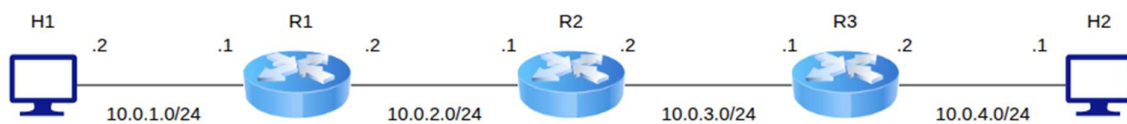
Every router initially only knows its directly connected networks. All routers support **equal-cost multi-path (ECMP)**: a router keeps every next hop that ties the lowest metric to a destination. Unless a question says otherwise, all routers run **split horizon with poison reverse**. Write each route as **destination, metric, next hop** using “-” for a directly connected network.

- a) After RIPv2 has run long enough that it converges, write down the complete routing table of router C. List all networks with their metric and next hop(s), and show every next hop where router C has more than one equally good choice. Answer with a table of the below format. (5 points)

Destination	Metric	Next hop(s)
192.168.1.0/24		
192.168.2.0/24		
192.168.3.0/24		
192.168.4.0/24		
192.168.5.0/24		

- b) (i) After convergence, router C has to forward two packets: one addressed to 192.168.1.88 and one addressed to 192.168.5.140. For each packet, state which next hop(s) C is allowed to use, and explain in one or two sentences why C may split traffic across two paths for one destination but not the other (3 points).
- (ii) State one benefit of splitting traffic between multiple paths (1 point).
- c) At some point the link that connects Router A to net5 fails. Assume the routers exchange their full distance vectors in **synchronized rounds** (in each round, every router simultaneously sends its current distance vector table, then all routers process what they received).
- For this part only, assume the routers run RIPv2 **with split horizon and poison reverse switched off**. Starting from the converged state, explain what happens to the metric that routers A, B and C hold for **net5** for a few (3-4) rounds. Can you see a pattern? After how many rounds will the network converge? (3 points)
 - Now assume split horizon and poison reverse are switched **on**. Show that the same failure now converges in a fewer number of rounds (how many?) and explain which specific loop poison reverse breaks to make this happen. (3 points)

4. ICMP (15/100)



Hosts H1 and H2 are connected to an IPv4 network, as shown above. All forwarding tables are correctly configured. The IP addresses of the hosts and the router interfaces are as shown in the figure, the link MTUs are provided in the following table.

Network	Address Block	MTU
net1	10.0.1.0/24	1500
net2	10.0.2.0/24	1500
net3	10.0.3.0/24	620
net4	10.0.4.0/24	1500

- a) H1 runs *traceroute* to 10.0.4.1 (H2) using ICMP Echo probes, sending exactly one probe per hop. List every IP datagram carrying an ICMP message that **H1** receives, and every

- that **H2** receives, in order. For each datagram, give the source IP, the destination IP, and the ICMP type and code. (5 points)
- b) An application on **H2** sends 1000 bytes of data using UDP to **H1**. The **datagram** carrying the data has the Don't Fragment bit set to 1. Will H2 receive an ICMP error message? If so, give the type and code, and the source and destination IP of the ICMP message. (5 points)
 - c) An ICMP message arrives carrying the following ICMP header (in hexadecimal):

03 03 A2 1C 00 00 00 00

What is the type? What is the code? What does the message mean, and in what situation is it sent? Can you verify from the bytes shown whether the checksum is correct? (5 points)

5. TCP (20/100)

Consider a recently established TCP connection between processes P_A and P_B on hosts A and B , respectively. The three-way handshake has been completed, but no data has been sent yet. TCP on Host B announced a receiver window size of 3600 bytes to TCP on Host A , and Process P_B can read the received data from TCP as soon as they arrive. Process P_A has 13200 bytes to send via TCP. The path MTU between the two hosts is known to be 1240 bytes. The one-way propagation time is 50ms, and the link speed is 960 Kbps.

The receiver uses delayed acknowledgments with a delay of 300ms (or at most two full sized segments). We consider that the size of a segment having a TCP header only is negligible in terms of transmission time. Moreover, for segments containing TCP payload, the size of the headers can be neglected when computing the transmission time of the segment. IPv4 is used as the network layer protocol, and IP options are not used. The CWND is originally set to 1 MSS and the slow start threshold is 65535 bytes. Assume that the granularity G of the heartbeat timer is 0.25 seconds.

Process P_A sends the first segment at time $t_0 = 0$ with sequence number 1000. The **second acknowledgement segment** that the receiver sends after t_0 is lost. Consequently, the sender carries out a retransmission at time t_1 . Assume that there is no other packet loss or retransmission during the entire connection.

- a) Find the value of t_1 . (7p)
- b) Provide the sequence of segments sent by TCP from host A . For each segment sent from host A provide the time it is sent, and the sequence number of the first byte it contains. Assume that outgoing segments are handled before incoming segments in case more than one event happens at the same time! (10p)
- c) At what time does A receive the acknowledgement for the last segment? (3p)

Hint 1: Try to first draw the sequence of segment exchanges to get the order of the segments right.

Hint 2: Consult RFC 6298. The description provided in the course book (3ed and 4ed) about retransmission time is not correct. Consult RFC5681 for congestion control.

6. UDP (15/100)

Process P_A on host A wants to send a message containing the word CLOUD to a process P_B on host B via UDP using ASCII character encoding. The data are transmitted over an IPv4 network, and IP options are not used. The IP addresses of hosts A and B are 198.53.2.10 and 198.51.101.20, respectively. The UDP header of the datagram received by B in hexadecimal format is as follows:

C350 1F90 000D X,

where X represents 2 bytes (4 hexadecimal digits).

Determine all possible values of X for which the UDP datagram will not be dropped by the UDP receiver. Motivate your answer.

Hint: For a reference to ASCII codes, you could use [ASCII table - Table of ASCII codes, characters and symbols](#).